Computational-Basis Success Probability under Phase-Flip Noise

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Abstract

In this study, we investigated whether computational-basis success probability alone can correctly evaluate a phase-corrupted Bell-state circuit. The ideal state $|\Phi^+\rangle = (|00\rangle + |11\rangle)/\sqrt{2}$ was subjected to a phase-flip channel on the first qubit, and the error probability was swept from $p = 0$ to $p = 1$ in steps of 0.01. The computational-basis success probability, fidelity to $|\Phi^+\rangle$, and the signed and absolute values of the coherence element $\rho_{00,11}$ were compared. The success probability remained equal to one for all 101 simulated values, while the fidelity decreased from one to zero and $\mathrm{Re}(\rho_{00,11})$ changed from 0.5 to -0.5. At $p = 0.5$, the coherence vanished even though the selected success metric still reported a perfect result. Therefore, computational-basis success probability alone cannot certify the relative phase of the prepared Bell state.

Introduction

Noise limits the reliability of near-term quantum circuits and makes the choice of evaluation metric important [2]. A metric may correctly measure one property of a state while remaining insensitive to another. For the Bell state $|\Phi^+\rangle$, measurements in the computational basis produce 00 and 11 with equal probabilities. It is therefore natural to define both outcomes as successful and use $P(00) + P(11)$ to evaluate the circuit.

However, the same computational-basis outcomes are obtained from $|\Phi^-\rangle = (|00\rangle - |11\rangle)/\sqrt{2}$ and from the incoherent mixture $\frac{1}{2}|00\rangle\langle 00| + \frac{1}{2}|11\rangle\langle 11|$. These states have different relative phases or coherence even though their populations are identical. State fidelity and density-matrix elements contain information that is not present in the selected success probability [1]. In this work, a phase-flip channel represented by Kraus operators was used to isolate this difference [3]. The purpose was to determine whether a circuit can be judged as successful after the intended Bell state has been changed.

Methodology

Bell-state preparation

The computational basis was ordered as $(|00\rangle, |01\rangle, |10\rangle, |11\rangle)$. Beginning with $|00\rangle$, a Hadamard gate was applied to the first qubit:

$$(H \otimes I)|00\rangle = H|0\rangle \otimes I|0\rangle \tag{1}$$

$$= \frac{|00\rangle + |10\rangle}{\sqrt{2}}. \tag{2}$$

A CNOT gate with the first qubit as the control transforms $|10\rangle \to |11\rangle$, and the output is

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2}}\begin{pmatrix}1 & 0 & 0 & 1\end{pmatrix}^T. \tag{3}$$

The density matrix is obtained from the outer product:

$$\rho_+ = |\Phi^+\rangle\langle\Phi^+| \tag{4}$$

$$= \frac{1}{2}(|00\rangle + |11\rangle)(\langle 00| + \langle 11|) \tag{5}$$

$$= \frac{1}{2}(|00\rangle\langle 00| + |00\rangle\langle 11| + |11\rangle\langle 00| + |11\rangle\langle 11|) \tag{6}$$

$$= \frac{1}{2}\begin{pmatrix}1 & 0 & 0 & 1\\ 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0\\ 1 & 0 & 0 & 1\end{pmatrix}. \tag{7}$$

Thus, the population elements are $\rho_{00,00} = \rho_{11,11} = 1/2$, while the off-diagonal coherence elements are $\rho_{00,11} = \rho_{11,00} = 1/2$.

Phase-flip channel

The phase flip was applied to the first qubit using

$$Z_1 = Z \otimes I = \mathrm{diag}(1, 1, -1, -1). \tag{8}$$

For an error probability p, the channel is

$$\mathcal{E}_p(\rho) = (1 - p)\rho + pZ_1\rho Z_1, \tag{9}$$

with Kraus operators

$$E_0 = \sqrt{1 - p}\, I_4, \qquad E_1 = \sqrt{p}\, Z_1. \tag{10}$$

The completeness relation was verified explicitly:

$$E_0^\dagger E_0 + E_1^\dagger E_1 = (1 - p)I_4^\dagger I_4 + pZ_1^\dagger Z_1 \tag{11}$$

$$= (1 - p)I_4 + pZ_1^2 \tag{12}$$

$$= (1 - p)I_4 + pI_4 = I_4. \tag{13}$$

Therefore, the map is trace preserving. Its operator-sum representation also guarantees complete positivity [3].

Since

$$Z_1|\Phi^+\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}} = |\Phi^-\rangle, \tag{14}$$

the noisy state is

$$\rho(p) = (1 - p)|\Phi^+\rangle\langle\Phi^+| + p|\Phi^-\rangle\langle\Phi^-|. \tag{15}$$

Expanding the two projectors gives

$$\rho(p) = \frac{1}{2}\begin{pmatrix} 1 & 0 & 0 & 1-2p \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 1-2p & 0 & 0 & 1 \end{pmatrix}. \tag{16}$$

For additional verification, let the diagonal entries of Z_1 be $(z_{00}, z_{01}, z_{10}, z_{11}) = (1, 1, -1, -1)$. Because $(Z_1\rho Z_1)_{ij} = z_i z_j \rho_{ij}$, the relevant density-matrix indices are

$$\rho_{00,00}(p) = (1-p)\frac{1}{2} + p(1)(1)\frac{1}{2} = \frac{1}{2}, \tag{17}$$

$$\rho_{11,11}(p) = (1-p)\frac{1}{2} + p(-1)(-1)\frac{1}{2} = \frac{1}{2}, \tag{18}$$

$$\rho_{00,11}(p) = (1-p)\frac{1}{2} + p(1)(-1)\frac{1}{2} = \frac{1-2p}{2}, \tag{19}$$

$$\rho_{11,00}(p) = \frac{1-2p}{2}. \tag{20}$$

Evaluation metrics

Three types of information were compared. First, the computational-basis success probability reads only the diagonal populations:

$$S(p) = P(00) + P(11) = \rho_{00,00}(p) + \rho_{11,11}(p) = 1. \tag{21}$$

Second, because the reference state is pure, its fidelity is

$$F(p) = \langle\Phi^+|\,\rho(p)\,|\Phi^+\rangle \tag{22}$$

$$= (1-p)|\,\langle\Phi^+|\Phi^+\rangle\,|^2 + p|\,\langle\Phi^+|\Phi^-\rangle\,|^2 \tag{23}$$

$$= 1 - p. \tag{24}$$

Third, the coherence was evaluated using both its signed real part and its magnitude:

$$\mathrm{Re}(\rho_{00,11}) = \frac{1-2p}{2}, \qquad |\rho_{00,11}| = \frac{|1-2p|}{2}. \tag{25}$$

The signed value distinguishes a positive from a negative relative phase, whereas the magnitude only indicates the amount of coherence.

Experiment Setup

The calculation was implemented in Python using NumPy. The phase-flip probability was swept over

$$p \in \{0, 0.01, 0.02, \ldots, 0.99, 1\}, \tag{26}$$

giving 101 deterministic density-matrix evaluations. For each value of p, the program applied the Kraus map and recorded $S(p)$, $F(p)$, $\mathrm{Re}(\rho_{00,11})$, and $|\rho_{00,11}|$. Matplotlib was used to generate the figures, and the complete numerical sweep was stored in CSV format. No random sampling, finite-shot measurement, or quantum software development kit was used.

The implementation was checked using nine pytest tests. These tests verify the Kraus completeness relation, unit trace, Hermiticity, positive semidefiniteness, the analytic end points, removal of coherence at $p = 0.5$, and the predicted behavior of the success probability and fidelity.

Results

Representative numerical values from the 101-point sweep are shown in Table 1. The simulated values agree with Eqs. (21)–(25) to the stored precision.

Table 1: The success probability remains one even when fidelity and signed coherence show that the target Bell state has changed.

p	$S(p)$	$F(p)$	$\mathrm{Re}(\rho_{00,11})$	$\lvert\rho_{00,11}\rvert$
0.00	1.00	1.00	0.50	0.50
0.25	1.00	0.75	0.25	0.25
0.50	1.00	0.50	0.00	0.00
0.75	1.00	0.25	-0.25	0.25
1.00	1.00	0.00	-0.50	0.50

Figure 1 compares the success probability with the target-state fidelity. The success probability remains equal to one over the entire interval, while the fidelity decreases linearly as the phase-flip probability increases.

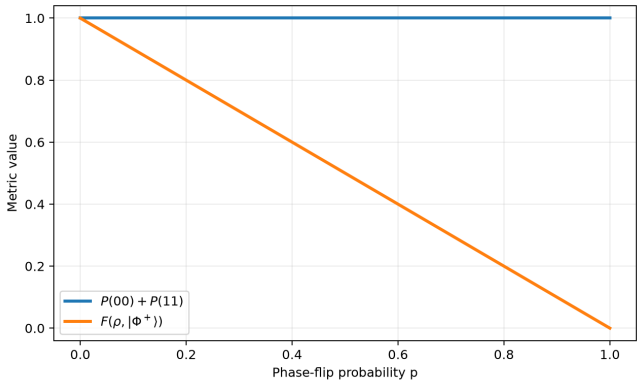


Figure 1: Computational-basis success remains perfect while fidelity decreases from one to zero; therefore, the accepted outcomes do not certify the intended Bell state.

At $p = 0.5$, Eq. (16) becomes

$$\rho(0.5) = \frac{1}{2}\,|00\rangle\langle 00| + \frac{1}{2}\,|11\rangle\langle 11|\,. \tag{27}$$

The off-diagonal coherence has vanished, but $S(0.5) = 1$. At $p = 1$, the state becomes $|\Phi^-\rangle\langle\Phi^-|$. It is fully coherent but orthogonal to the target, so $F(1) = 0$ while $S(1) = 1$.

Figure 2 shows why coherence magnitude alone is also insufficient. The magnitude decreases to zero at $p = 0.5$ and returns to 0.5 at $p = 1$, whereas the signed real part continues to -0.5 and exposes the phase reversal.

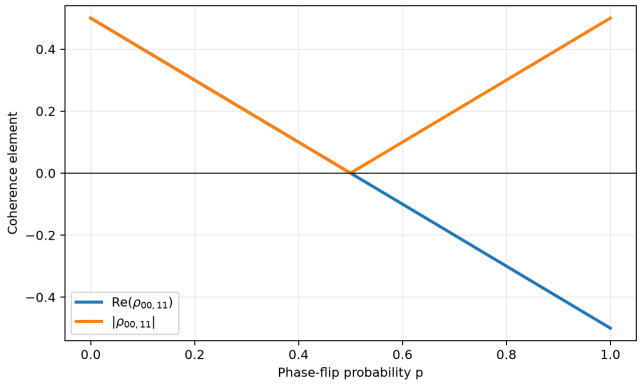


Figure 2: Signed coherence identifies the phase reversal at $p > 0.5$, while coherence magnitude alone cannot distinguish $|\Phi^+\rangle$ from $|\Phi^-\rangle$.

Conclusion

This experiment demonstrated that computational-basis success probability alone can incorrectly judge a phase-corrupted Bell-state circuit as perfectly successful. The value of $P(00) + P(11)$ remained equal to one for every phase-flip probability, including the incoherent state at $p = 0.5$ and the orthogonal Bell state $|\Phi^-\rangle$ at $p = 1$. In contrast, fidelity and the signed coherence element detected the change in the quantum state. Therefore, a benchmark intended to verify Bell-state preparation must include phase-sensitive information rather than relying only on accepted computational-basis outcomes.

Discussion

The present work is an analytic density-matrix calculation reproduced by a deterministic classical simulation. It does not contain data from quantum hardware. The phase-flip channel is a simplified noise model and does not include state-preparation-and-measurement (SPAM) error, crosstalk, coherent calibration error, finite-shot uncertainty, or drift. In addition, the noise channel was applied after an ideal Bell-state circuit and was not assigned to a specific pulse or time location inside a physical CNOT implementation. Consequently, the results do not estimate the CNOT error rate of IBM hardware or of any other real device.

The compared metrics also have different limitations. Fidelity determines whether the state agrees with one specified target, but it is not a complete characterization of a general quantum process. Coherence magnitude detects the loss and recovery of coherence but removes its sign, so it cannot distinguish $|\Phi^+\rangle$ from $|\Phi^-\rangle$. A future experiment may add finite-shot sampling and measurements in the X basis, compare phase-flip, bit-flip, and depolarizing channels, and then evaluate the same metrics using a noise model tied to a specified circuit location. Hardware measurements would require separate calibration data and should not be interpreted using the present simplified model alone.

References

[1] M. A. Nielsen and I. L. Chuang. *Quantum Computation and Quantum Information*. Cambridge University Press, 10th anniversary edition, 2010.

[2] J. Preskill. Quantum computing in the NISQ era and beyond. *Quantum*, 2:79, 2018.

[3] C. J. Wood, J. D. Biamonte, and D. G. Cory. Tensor networks and graphical calculus for open quantum systems. *Quantum Information and Computation*, 15(9–10):759–811, 2015.